

BELLS UNIVERSITY OF TECHNOLOGY
OTA, OGUN STATE, NIGERIA



COLLEGE OF NATURAL AND APPLIED SCIENCES
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION
TECHNOLOGY

Course Code: CSC 207 – Computer Logic & Digital Circuits Design
(3 Units)

2022/2023 FIRST SEMESTER EXAMINATION

Read the following instructions carefully

- Attempt Question One and any other Four (4) Questions
- Other than Question One, all other Questions carry equal marks
- Clear schematic illustrations, where required/ necessary also carry marks

TIME ALLOWED: 2.5 Hours

Question One (20 marks)

- a) Consider the decimal number 42.51 and do the following conversions: (6 marks)
 - i) decimal to binary
 - ii) decimal to octal
 - iii) decimal to hex-decimal
- b) Consider the binary number 1100110.111101 and do the following conversions (4 marks)
 - i) Binary to octal
 - ii) Binary to hexadecimal
- c) Define the following terms: (4 marks)
 - signed numbers
 - unsigned numbers
- d) Draw a 3 variable K-map (6 marks)

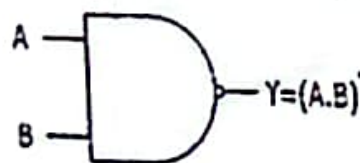
Question Two (12.5 marks)

- a) Define a logic gate. (2 marks)
- b) List and explain the 3 categories of logic gates. (4 ½ marks)
- c) Draw the symbol and the truth table of a 2-input AND, OR and NOT gate. (6 marks)

Question Three (12.5 marks)

- a) The symbol below represents which logic gate? Represent the symbol using a truth table. (6 marks)

A	B	$Y = (A \cdot B)$
0	0	
0	1	
1	0	
1	1	



Basic Logic: And, Or, Not
universal: NAND and NOR
special gate

b) i) Define the term 'encoder'. (2 marks)

ii) With the aid of a diagram, explain the 4 to 2 encoder. (8 marks)

Question Four (12.5 marks)

a) i) Define the term 'decoder'. (2 marks)

ii) With the aid of a diagram, explain 2 to 4 decoder (8 marks)

b) Explain the 2-level logic realization and their possible combinations (7 marks)

Question Five (12.5 marks)

a) i) What are 'multiplexers' (2.5 marks)

ii) With the aid of a diagram, explain the 4 X 1 multiplexer (4 marks)

b) i) What are 'de-multiplexers' (2 marks)

ii) With the aid of a diagram, explain the 1 X 4 de-multiplexer (4 marks)

Question Six (12.5 marks)

✓ (a) i) Define the term Boolean Algebra (3.5 marks)

✓ ii) State the three (3) basic laws of Boolean algebra (6 marks)

✓ (b) State the DeMorgan's theorem (3 marks)

$$Q = (\overline{A \cdot B}) \cdot (\overline{A \cdot B}) \cdot C$$

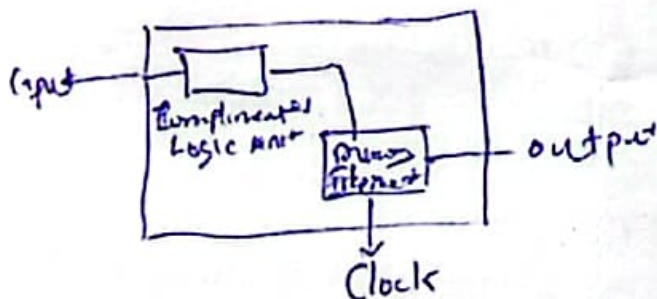
Question Seven (12.5 marks)

a) Write short notes on the following: (7.5 marks)

- i. Flip flops ii. Latches iii. Combinational circuit
- iv. Sequential circuit v. Counters

b) Discuss two (2) types of flip-flops showing each of their logic diagrams and truth table (5 marks)

This is a logic circuit which deals with the present input and past input



42.512

$$(2 \times 4^2) + (2 \times 2^2) = 1$$